

Geometrically Inductive Self-Supervised Learning for EEG: A Design Study of Spatial Attention Across Sessions, Subjects, and Montages

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Abstract

I propose, implement, and evaluate a self-supervised EEG encoder whose spatial attention is conditioned on a geometric descriptor g_{ij} — built from the 3-D scalp positions of each electrode pair — instead of a learned per-electrode codex. Because the attention function takes coordinates as input and holds no per-electrode parameters, the same trained encoder applies to any montage given only 3-D coordinates, including a montage never seen in pretraining. A codex, indexed by electrode identity, has no embedding for an unseen electrode and runs on a new montage only when the held-out channel names match pretraining ones (name-aligned transfer). The training recipe (momentum-encoder alignment with masked-patch reconstruction) follows EEGPT [14]. I test zero-shot cross-montage transfer with a leave-one-montage-out, mixed-corpus pretraining design. The only baseline that transfers zero-shot without modification is a channel-independent encoder (**chind**), which discards spatial structure. Pretrained on a balanced 156-subject corpus (PhysioNet MI + Sleep-EDFx) and evaluated zero-shot on the held-out 3-channel BCIC-2B montage, the geometric encoder reaches 0.545 BAC and beats **chind** (0.509) by a statistically significant +0.036 (paired t -test $p = 0.023$, 95% CI [+0.006, +0.066]). The position-only descriptor also beats it significantly (+0.031, $p = 0.022$, 95% CI [+0.006, +0.056]). The transductive codex, by name-aligned transfer in its best case, reaches only 0.522, below both geometric variants. At a smaller balanced 9-subject-per-dataset scale the geometric encoder leads **chind** on all three held-out montages, but the gaps are within noise. This is a design study: geometric conditioning delivers a significant zero-shot transfer advantage once the pretraining corpus is large enough, on the one held-out montage the three available datasets allow to be tested at scale.

Code: github.com/tianxin-scu/geometric-ee-ssl.

1 Introduction

Non-invasive BCI and EEG. Brain-computer interfaces (BCIs) offer a path to communication and control for individuals with motor impairments. Invasive modalities such as ECoG and Utah arrays deliver the highest signal fidelity but require neurosurgery. Electroencephalography (EEG) is the natural non-invasive alternative — safe, affordable, and widely deployed — but its scalp signals are noisy, mixing many cortical sources through volume conduction and contaminated by muscle and ocular artifacts. Better representation learning on EEG extracts more usable signal from the non-invasive modality, lowering the threshold at which a patient must accept surgical risk for functional BCI performance.

EEG is non-stationary, by default. Any deployed EEG model faces a signal that shifts continuously along three axes, none of which is an experimental edge case: (i) **cross-session drift** — recordings of the same subject on different days differ because the cap sits differently each application (several-millimeter placement shifts are typical) and because the subject’s physiological state varies (alertness, fatigue, electrode impedance); (ii) **cross-subject anatomical variability** — head size, cortical folding, and skull thickness differ across individuals, so C3 on one subject does not record from the same generators as C3 on another; and (iii) **cross-montage layout differences** — clinical and consumer devices deploy 2 to 256 electrodes in heterogeneous layouts, and a model trained on a 64-channel research montage has no principled way to process a 3-channel headset without retraining. The placement, anatomical, and montage components of these shifts share a common structure: the geometry of the electrode array, relative to the underlying cortical sources, has changed. A model whose spatial representation is conditioned on geometry should absorb that change smoothly, where a model whose spatial structure is encoded in per-electrode parameters cannot. Physiological state is a separate non-geometric axis of session variability and is not addressed by the architectural change proposed here.

The shared blind spot of EEG foundation models. Recent EEG foundation models — BENDR [10], BIOT [15], LaBraM [8], EEGPT [14] — bring self-supervised pretraining and linear-probe evaluation to EEG with strong downstream results, but they fall into two camps that both fail to handle EEG’s natural non-stationarity. The first camp uses a learned *codex*: a table of per-electrode embeddings $\{c_i\}$ indexed by electrode identity, added to patch projections before any spatial mixing (EEGPT, LaBraM). The codex embedding for C3 is a single vector fit to the average C3 placement across training subjects. It cannot express that today’s cap has shifted 5 mm or that this subject’s head is larger than average, and an electrode whose name never appeared in pretraining has no codex entry at all, so such a montage cannot be processed without inventing the missing rows. This is the classical *transductive* setting [5]: parameters indexed by node identity, no generalization to unseen nodes without retraining. The second camp removes per-electrode structure entirely: BIOT [15] tokenizes each channel independently, robust to variable channel counts but discarding the 3-D scalp coordinates that every EEG system provides as side information. Neither camp uses geometry as an input.

Contributions. I implement and evaluate the architecture committed to in the project proposal, and I focus the evaluation squarely on the cross-montage transfer the proposal called the headline test. Concretely: (i) **an architecture** in which per-electrode codex embeddings are replaced by a shared function over g_{ij} inside the attention computation, with no per-electrode parameters and an optional masked-reconstruction head — making the encoder *inductive* over electrode sets, and therefore runnable on any montage with known coordinates; (ii) **a leave-one-montage-out, mixed-corpus pretraining protocol** that trains on two datasets and evaluates zero-shot on the held-out third, so cross-montage generalization is the only variable that changes across splits; (iii) **two baselines.** The primary paired comparison is against a channel-independent encoder (**chind**), the only baseline that transfers zero-shot without modification, trained under identical conditions. The transductive codex is also included through name-aligned transfer, which copies a held-out electrode’s embedding from a same-named pretraining electrode — the only way an identity-indexed codex can run on a new montage, and possible here because all three BCIC-2B channel names matched; (iv) **an injection-site and descriptor ablation:** I report the full injection-site comparison (G1/G2/G3) and ablate the descriptor content — the full g_{ij} against a position-only variant — taking G3 (geometry at both attention score and value) as the primary configuration because

it transfers significantly under both descriptors; and (v) **the headline result**. At sufficient pre-training scale the geometric encoder beats **chind** on zero-shot transfer by a statistically significant margin. At small scale the advantage is directionally consistent across all three held-out montages but not individually significant.

2 Problem Statement

Formal setup. Let $\mathbf{X} \in \mathbb{R}^{M \times T}$ denote a multichannel EEG recording with M electrodes and T time samples, and let $\mathbf{p}_i \in \mathbb{R}^3$ denote the known 3-D scalp position of electrode i , taken from the recording montage’s digitized coordinates or a standard template when digitization is unavailable. Self-supervised pretraining learns an encoder $f_{\theta} : \mathbb{R}^{M \times T} \rightarrow \mathbb{R}^d$ from unlabeled recordings. Pretraining succeeds if a linear classifier trained on top of f_{θ} transfers across diverse downstream tasks. This linear-probing protocol is the standard evaluation framework in EEG foundation-model work [8, 10, 14, 15], and I adopt it here.

What this project isolates. The open design choice I focus on is how the spatial encoder represents inter-electrode relationships. Existing foundation models split between codex-based spatial encoders (transductive in electrode identity) and channel-independent encoders (no spatial structure at all). I propose a third option: a spatial attention function conditioned on a geometric descriptor g_{ij} (Eq. 1) with no per-electrode parameters, so the same shared function applies to any electrode pair given only their 3-D positions. This makes the encoder inductive over electrode sets [5] and applicable, frozen, to a montage it never saw in pretraining.

Research question.

Does conditioning the spatial encoder’s attention on electrode geometry — rather than on a learned codex of electrode identities — produce representations that transfer zero-shot to an unseen electrode montage, beating the only baseline that can also transfer (a channel-independent encoder)?

This is a design study, not a SOTA competition. I do not ask whether my model outperforms LaBraM or EEGPT trained at scale. I ask whether, controlling for parameter count and pretraining data, geometric conditioning provides a cross-montage transfer advantage over the comparable baseline — and whether that advantage is large enough to be statistically real.

3 Related Work

3.1 EEG Foundation Models

Recent EEG foundation models — BENDR [10], BIOT [15], LaBraM [8], EEGPT [14] — share a pipeline of large-scale self-supervised pretraining followed by linear probing, and transfer across motor-imagery and clinical EEG tasks. A shared limitation remains: codex-based models encode space through per-electrode identities (transductive), while channel-independent models omit spatial structure entirely. Montage heterogeneity — 2 to 256 channels in varying layouts — is the deployment bottleneck this work addresses.

3.2 Self-Supervised Learning with Momentum Encoders

Momentum-encoder learning from MoCo [7] and BYOL [4] provides stable targets without hard-negative mining, and MAE [6] adds a reconstruction objective. I adopt this validated combination (alignment + optional reconstruction), as in EEGPT [14], and focus the novelty on the spatial representation rather than the objective.

3.3 Inductive Representations and Attention over Relational Structure

The framing — replacing identity-indexed parameters with a shared function over structural inputs — comes from the inductive node-embedding literature. GraphSAGE [5] distinguished *transductive* embeddings (fit per-node, fixed graph) from *inductive* aggregation functions (shared parameters over node and neighbourhood features) that generalize to unseen nodes, which is the distinction I draw between codex-based and geometry-conditioned spatial encoders. GAT [13] showed that attention conditioned on relational features is a natural such aggregator, the pattern I adopt inside the spatial Transformer. EEG connectivity studies independently document that task-relevant information is partly relational [2]. This is a Transformer with a geometry-conditioned attention bias, not a sparse graph or message passing, and full $M \times M$ attention is within budget at the electrode counts considered ($M \leq 64$).

4 Proposed Solution

4.1 Architecture Overview

Figure 1a summarizes the model. The single architectural choice that distinguishes this design from existing EEG foundation models lives inside the spatial encoder: how electrode-specific information enters the attention computation, illustrated in Figure 1b. All other components — patching, momentum updates, masking, temporal contextualization, and loss — follow standard practice [4, 6, 14]. The backbone recomputes geometry tables per dataset per step, because the montage — and therefore g_{ij} — changes between corpora within a single mixed-corpus pretraining run.

4.2 The Geometric Descriptor

For each ordered pair of electrodes (i, j) with 3-D scalp coordinates $p_i, p_j \in \mathbb{R}^3$, the headline descriptor is

$$g_{ij} = [p_i, p_j, p_i - p_j, \|p_i - p_j\|_2] \in \mathbb{R}^{10}. \quad (1)$$

It combines absolute positions (p_i, p_j) , signed displacement $(p_i - p_j, \text{distinguishing anterior/posterior and left/right})$, and distance (a scalar proximity term). Including the two absolute positions, beyond displacement and distance, lets the shared MLP localize a pair on the scalp rather than only relate the two electrodes. The descriptor is asymmetric in (i, j) , so attention can express directional preferences. Because g_{ij} is a function of coordinates alone and contains no electrode-identity parameters, the same descriptor space is defined for any montage with known scalp coordinates — including montages absent from pretraining.

Coordinates are computed in a fixed scale from each dataset’s channel names, with no per-montage normalization. This matters for cross-montage consistency: the encoder must see the held-out montage’s coordinates on the same scale it saw during pretraining, or the transfer is confounded.

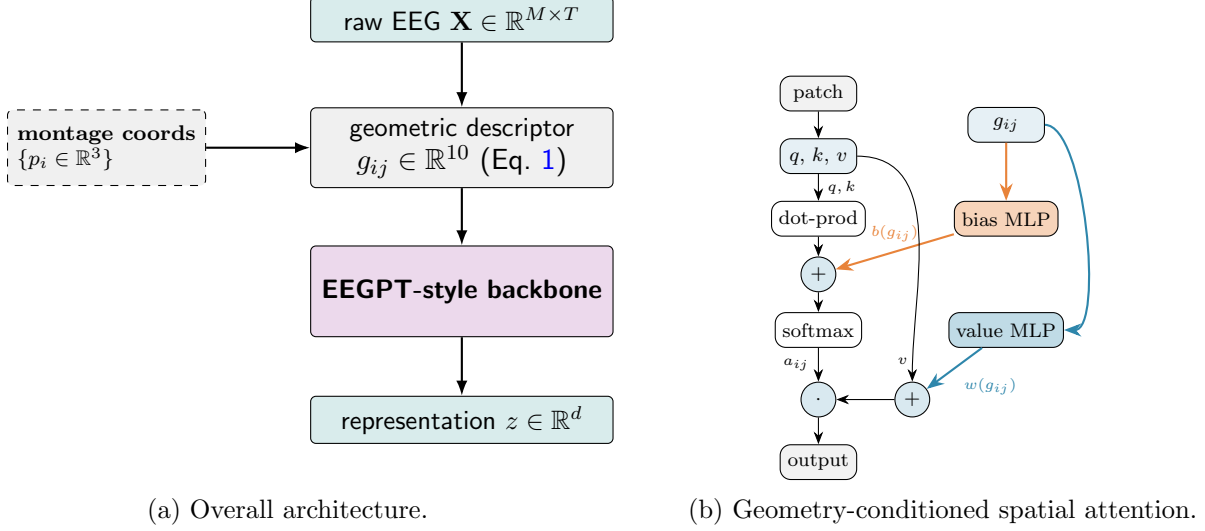


Figure 1: **Model overview.** (a) Raw EEG and the montage’s electrode coordinates yield a geometric descriptor g_{ij} that conditions the spatial attention of an otherwise standard EEGPT-style backbone. The per-electrode codex used by EEGPT is removed, and g_{ij} takes its place as the spatial inductive bias. Coordinates are side information read from the recording montage, digitized when available or taken from a 10–20 template. (b) Inside spatial attention, g_{ij} enters through two shared MLPs: a bias path added to the attention score (G1) and a value path added to each message (G2), with G3 using both. No per-electrode parameters appear, so the same function applies to any montage with known coordinates.

Descriptor ablation (position-only). I also evaluate a reduced descriptor $g_{ij} = [p_i, p_j] \in \mathbb{R}^6$ (absolute positions only, dropping the displacement and distance terms). This asks whether absolute position alone carries the cross-montage signal, or whether the relational displacement/distance terms are doing the work.

4.3 Geometric vs. Channel-Independent Spatial Attention

Writing the standard dot-product attention as $\alpha_{ij} \propto \exp((W_Q x_i)^\top (W_K x_j) / \sqrt{d})$ with output $y_i = \sum_j \alpha_{ij} W_V x_j$, geometry can enter at the score (G1), the value (G2), or both (G3). Figure 1b shows the two paths:

$$\text{G1: } e_{ij} = (W_Q x_i)^\top (W_K x_j) / \sqrt{d_h} + b_h(g_{ij}), \quad (2)$$

$$\text{G2: } v_{j \rightarrow i} = W_V x_j + W_{V,g} g_{ij}, \quad y_i = \sum_j \alpha_{ij} v_{j \rightarrow i}, \quad (3)$$

where b_h is a shared MLP $\mathbb{R}^{10} \rightarrow \mathbb{R}^H$ (one output per head). G1 conditions which electrodes attend, and G2 conditions the content of each message. G3 applies both with separate projections of the same g_{ij} . I take G3 as the primary configuration and ablate the descriptor content under it (Section 4.2). The full injection-site ablation (Section 7.3) supports this: G3 transfers significantly under both the full and position-only descriptors and is the strongest injection on average, while G2 gives the single highest cell but only with the full descriptor.

Channel-independent baseline (chind). The same backbone with no geometric term and no per-electrode parameters: standard dot-product attention only. Because **chind** carries no electrode-

identity parameters either, it too can be applied to an unseen montage, so it is the appropriate baseline for a zero-shot cross-montage test.

4.4 Self-Supervised Objective

The encoder is supervised by the combination of momentum-encoder alignment and masked-patch reconstruction:

$$\mathcal{L} = \mathcal{L}_A + \lambda_R \mathcal{L}_R. \quad (4)$$

$\mathcal{L}_A$ is the alignment term, $-2 \cdot \cos(\text{pred}(z_{\text{online}}), \text{stopgrad}(z_{\text{momentum}}))$ averaged over batch and time. $\mathcal{L}_R$ is masked-patch reconstruction MSE, applied only at masked time steps, with a per-channel reconstruction target. I default to $\lambda_R = 1$ with 50% temporal masking (temporal-only, no spatial masking). The momentum encoder is a non-trainable EMA copy of the online encoder, processing the unmasked input. The EMA factor follows a cosine ramp $\tau(k) = 1 - (1 - \tau_0) \frac{\cos(\pi k/K) + 1}{2}$ with $\tau_0 = 0.996$ and $\tau_K = 1.0$, matching the EEGPT recipe.

4.5 Baselines: Channel-Independent and Name-Aligned Codex

The primary comparison baseline is the channel-independent encoder of Section 4.3, trained under identical conditions. It carries no per-electrode parameters, so it transfers zero-shot without modification and supports a paired significance test against the geometric encoder.

The transductive codex is also reported, through *name-aligned transfer*. A per-electrode codex has no embedding for a held-out electrode whose name was absent from pretraining. Name-aligned transfer copies the codex row of each same-named pretraining electrode, falling back (e.g. random initialization) only for unmatched names. For the held-out BCIC-2B montage all three channels (C3, Cz, C4) matched the pretraining montage, so every electrode received its fully trained embedding and no fallback was triggered — the codex’s best case. Because the copy is keyed by channel name rather than by coordinates, it does not reintroduce geometry at transfer time, so it is a clean codex point estimate rather than a geometry-in-disguise baseline.

One caveat makes this comparison conservative: the codex checkpoint was pretrained on PhysioNet MI alone, not the mixed 156-subject corpus used for the geometric and channel-independent encoders. That regime favors the codex — more in-domain data and a fully matched montage — so a codex that still underperforms geometry is a comparison biased toward the codex. I report it as a point estimate with no paired test. A mixed-corpus codex pretrain is the clean follow-up (Section 10).

5 Experiment Design

The experiment is a single design: leave-one-montage-out, mixed-corpus pretraining, with zero-shot linear-probe evaluation on the held-out montage. This isolates cross-montage transfer — the proposal’s headline promise — as the dependent variable, and drops the in-distribution and cross-session ladder rungs of an earlier exploratory regime (which had shown no separation and are not re-run here).

Variants. The full geometric variant (G3, full 10-D g_{ij}), the position-only variant (G3, 6-D descriptor $[p_i, p_j]$), and the channel-independent baseline **chind** (no geometry). Every variant is trained under identical conditions — same architecture, optimizer, schedule, masking, loss, and EMA schedule — so only the spatial encoder differs.

Table 1: Leave-one-montage-out splits. All three run at the balanced n9 scale (Table 3). Only the BCIC-2B hold-out split can be scaled to n156 (Table 4), because any balanced corpus containing BCIC-2B is capped at 9 subjects (Section 6.1).

Pretrain on	Held out (zero-shot eval)	Scales
PhysioNet MI + BCIC-2B	Sleep-EDFx	n9
PhysioNet MI + Sleep-EDFx	BCIC-2B	n9, n156
BCIC-2B + Sleep-EDFx	PhysioNet MI	n9

Two scales. (1) **Balanced n9.** Subjects capped at 9 per dataset (BCIC-2B has only 9 total — that hard ceiling sets the shared budget, so every dataset contributes equal subject-level diversity). All three splits $\times$ three variants give the directional table in Section 7.1. (2) **Scaled n156.** The BCIC-2B hold-out split scaled to 78 subjects each from PhysioNet MI and Sleep-EDFx (a balanced 156-subject corpus), evaluated zero-shot on BCIC-2B. This is the only split that can be both balanced and large (Section 6.1), and is the powered headline test (Section 7.2).

The pipeline has three stages: mixed-corpus pretraining per (variant, split), a frozen-backbone zero-shot linear probe on the held-out montage, and aggregation of the per-checkpoint results into the headline tables.

6 Experimental Setup and Datasets

6.1 Datasets

I use three open EEG datasets accessible through MNE-Python [3] and MOABB [1].

Table 2: Open EEG datasets used in this project.

Dataset	Paradigm	Subj.	Ch.	Rate	Role
PhysioNet MI [11]	Motor imagery (4-class)	105	64	160 Hz	pretrain / held-out
BCIC-2B [12]	Motor imagery (binary)	9	3	250 Hz	pretrain / held-out
Sleep-EDFx [9]	Sleep staging	78	2 bipolar	100 Hz	pretrain / held-out

Four PhysioNet MI subjects (88, 92, 100, 104) are excluded for known annotation/recording inconsistencies, leaving 105 usable.

The 9-subject ceiling, and why it shapes the whole design. BCIC-2B has only 9 subjects total. This single fact determines what experiments are possible: (a) any balanced pretraining corpus that includes BCIC-2B is capped at 9 subjects per dataset; (b) therefore the only corpus that can be both balanced and large (n156) is the one that holds BCIC-2B out; and (c) BCIC-2B’s 3-channel layout is a sparse montage, far from the 64-channel PhysioNet cap the model must transfer from, which makes it a demanding zero-shot target. The 9-subject ceiling is thus simultaneously the reason the powered test is possible and the reason there is only one of it.

BCIC-2A is reserved as future work. 2A is 22-channel 4-class motor imagery on 9 subjects — a third in-distribution-style motor-imagery dataset rather than a new perturbation axis, and equally subject-capped. Adding it (or another distinct-montage dataset) as a second held-out test is the most valuable follow-up (Section 10).

6.2 Preprocessing (Locked Across Datasets)

Identical preprocessing is applied to all three datasets so that cross-dataset comparisons isolate dataset characteristics, not preprocessing decisions:

- Bandpass 0.5–45 Hz, zero-phase FIR.
- Resample to 200 Hz (PhysioNet 160 $\rightarrow$ up, BCIC 250 $\rightarrow$ down, Sleep-EDFx 100 $\rightarrow$ up, all end identical).
- Epoch length 4 s (16 patches at 250 ms per epoch).
- Per-epoch linear detrend, per-epoch per-channel z-score.
- No baseline correction, no re-referencing, no ICA.
- Standard 10–05 montage, fixed-scale coordinates (no per-montage normalization, Section 4.2).

Sleep-EDFx specifics. The dataset uses two bipolar derivations (Fpz--Cz, Pz--Oz) not in any standard montage. I place each bipolar electrode at the midpoint of its reference pair in standard 10–05 coordinates. Sleep-EDFx is annotated with variable-length sleep-stage spans. I slice each span into non-overlapping 4 s sub-epochs inheriting its stage label, to keep the backbone unchanged.

6.3 Architecture Defaults

Token dimension $d = 256$; spatial depth $L_S = 2$; temporal depth $L_T = 4$; 8 heads; MLP ratio 4. Patch length 50 samples (250 ms at 200 Hz), non-overlapping; 16 patches per 4 s epoch. Geometric MLP hidden width 32. Temporal positional embedding: learned.

The geometric and channel-independent backbones share an identical temporal-Transformer / projection / patcher stack (~ 4.8 M online parameters in the backbone). Only the spatial encoder differs: the geometric variants add a small shared MLP over g_{ij} (no per-electrode parameters), and **chind** adds nothing. Because neither variant carries per-electrode parameters, both are montage-agnostic and can be evaluated zero-shot.

6.4 Training

AdamW (lr $1 \cdot 10^{-4}$, weight decay 0.01); linear warmup for 10 epochs then cosine decay over 100 total epochs; EMA τ cosine ramp $0.996 \rightarrow 1.0$; batch size 64; $\lambda_R = 1$ with 50% temporal masking; epoch budget 4096 per dataset per pass (small datasets oversampled, large ones subsampled to this exact count, equalizing gradient weight across the mixed corpus). Geometry tables are recomputed each training step (the geometric MLPs are trainable, and the montage changes between corpora within a run). At inference there is no backward pass, so tables are precomputed once per montage.

6.5 Compute

Pretraining runs on Google Colab (T4 / A100) and on a local NVIDIA RTX 4070. A balanced n156 pretraining run (100 epochs) completes in ~ 40 min on the RTX 4070, and the zero-shot probe adds ~ 2 min. Every checkpoint records its resolved preprocessing, ablation, and training settings alongside the variant, pretrain datasets, held-out dataset, epoch budget, and subject count, so evaluation never relies on source-code defaults.

6.6 Linear-Probe Protocol

For each (variant, split) checkpoint I load the pretrained online backbone and freeze all parameters, then recompute the held-out montage’s fixed-scale coordinates from its channel names. I extract a fixed feature per epoch by mean-pooling the backbone output over electrodes and time to obtain a single $\mathbb{R}^{256}$ vector, standardize on the training split only (no leakage), and fit a strictly linear logistic regression (no MLP probe), so the result reflects representation quality rather than probe capacity. Each held-out dataset uses 9 subjects for the balanced table (for cross-row comparability): LOSO for PhysioNet MI and BCIC-2B, and a within-subject night split for Sleep-EDFx (train on night 1, test on night 2). The whole held-out dataset is excluded from pretraining, so every evaluated subject is genuinely zero-shot.

6.7 Significance Testing

For the headline $n156 \rightarrow \text{BCIC}$ test I run a paired t -test on per-subject BAC differences between a geometric variant and `chind` over the 9 LOSO folds (the same held-out subjects for both models) at $\alpha = 0.05$, and report the 95% confidence interval of the mean difference.

7 Results

All BAC values are LOSO or within-subject-night means $\pm$ standard deviations across the evaluated subjects.

7.1 Balanced n9 — Directional Table Across All Three Held-Out Montages

Table 3: Balanced 9-subject-per-dataset corpus. Zero-shot BAC on each held-out montage. Read within each row (geometric vs. `chind`), as rows are different tasks with different chance levels.

Held-out (eval)	geometric (full)	geometric (position-only)	chind
Sleep-EDFx (night split)	0.593 $\pm$ 0.022	0.597 $\pm$ 0.022	0.589 $\pm$ 0.020
BCIC-2B (LOSO)	0.509 $\pm$ 0.026	0.508 $\pm$ 0.019	0.504 $\pm$ 0.024
PhysioNet MI (LOSO)	0.303 $\pm$ 0.048	0.315 $\pm$ 0.044	0.302 $\pm$ 0.047

With a balanced 9-subject-per-dataset corpus, both geometric variants beat `chind` on all three held-out montages. The direction is consistent, but the gaps are small (≤ 1.3 BAC points) and within the per-subject standard deviation, so no single row is significant on its own. The three rows are different tasks with different chance levels (BCIC-2B binary, chance 0.50; PhysioNet MI 4-class, chance 0.25; Sleep-EDFx multi-class), so the comparison is within each row, not across rows. This table establishes a consistent directional signal and motivates scaling the one split that can be scaled.

7.2 Scaled n156 — The Powered Zero-Shot Transfer Test (Held-Out BCIC-2B)

Pretraining on a balanced 156-subject corpus (PhysioNet MI 78 + Sleep-EDFx 78), evaluated zero-shot on BCIC-2B (9 LOSO folds):

When the pretraining corpus is large enough, the geometric advantage over the baseline becomes both larger and statistically significant for both descriptors. The full 10-D descriptor gives the strongest point estimate (0.545, +0.036 over `chind`). The reduced position-only descriptor still

Table 4: Scaled n156 run: zero-shot BAC on the held-out BCIC-2B montage. The codex is a point estimate via name-aligned transfer from a PhysioNet-only checkpoint (Section 4.5), not a mixed-corpus run, so it carries no paired test.

Variant	BCIC-2B zero-shot BAC (n=9)
chind (baseline)	0.509 $\pm$ 0.016
codex (name-aligned transfer)	0.522 $\pm$ 0.021
geometric (position-only)	0.540 $\pm$ 0.026
geometric (full g_{ij})	0.545 $\pm$ 0.027

Table 5: Paired t -test vs. **chind** over the same 9 held-out subjects (each model on the identical LOSO folds), significant at $\alpha = 0.05$ for both descriptors, with both confidence intervals excluding zero.

Comparison	paired t	95% CI
geometric (full) – chind	$p = 0.023$	[+0.006, +0.066]
geometric (position-only) – chind	$p = 0.022$	[+0.006, +0.056]

beats **chind** significantly (+0.031), so the cross-montage signal survives dropping the displacement and distance terms. The name-aligned codex reaches 0.522: above **chind** but below both geometric variants, even though its PhysioNet-only checkpoint and fully name-matched montage make this its most favorable case (Section 4.5). The same comparison at the balanced n9 scale (Section 7.1) showed only +0.003 BAC (not significant), so the effect emerges with pretraining scale.

7.3 Injection-Site Ablation (E5)

Table 6 crosses the three injection sites (G1 score bias, G2 value, G3 both) with the two descriptors at the n156 scale on the held-out BCIC-2B montage. The value path carries the cross-montage signal: G2 and G3 with the full descriptor both beat **chind** at $\alpha = 0.05$, while G1 alone reaches 0.510 ($p = 0.854$), within noise of the baseline, so a score bias over g_{ij} adds almost nothing when the value path is absent. G2 with the full descriptor gives the single highest cell (0.549), but its advantage holds only there: under the position-only descriptor G2 falls to +0.011 ($p = 0.229$), not significant. G3 is the more robust injection. Both G3 descriptors beat **chind** significantly (full +0.036, $p = 0.023$; position-only +0.031, $p = 0.022$), and averaged over the two descriptors G3 leads G2 (0.543 vs 0.535). I therefore report G3 as the primary configuration and note G2 with the full descriptor as the single strongest result. The displacement and distance terms matter more when geometry enters through the value path alone: dropping them costs G2 its significance but leaves G3 significant.

8 Discussion

8.1 What the Results Say

The experiments support the proposal’s central claim on the test they can support best. Geometric conditioning produces representations that transfer zero-shot to an unseen montage and beat the channel-independent baseline — the only one that transfers without modification — by a margin that survives a paired significance test once the pretraining corpus is large enough (n156). They

Table 6: Injection-site ablation at n156 (zero-shot BCIC-2B, 9 LOSO folds). The three injection sites crossed with the two descriptors. The win count, paired t -test p -value, and 95% CI are computed against `chind` over the same 9 held-out subjects. G3 is the primary configuration, significant under both descriptors and strongest on average. G2 with the full descriptor gives the single highest cell.

Cell	BAC	wins	paired t	95% CI
<code>chind</code> (base)	0.509	—	—	—
G1 + full	0.510	5/9	$p = 0.854$	$[-0.008, +0.010]$
G1 + position-only	0.506	4/9	$p = 0.402$	$[-0.010, +0.004]$
G2 + full	0.549	8/9	$p = 0.004$	$[+0.017, +0.063]$
G2 + position-only	0.520	5/9	$p = 0.229$	$[-0.008, +0.030]$
G3 + full	0.545	7/9	$p = 0.023$	$[+0.006, +0.066]$
G3 + position-only	0.540	7/9	$p = 0.022$	$[+0.006, +0.056]$

also beat the name-aligned codex, whose best case still trails both geometric variants. At small balanced scale (n9) the advantage is consistent in direction across all three held-out montages but not in magnitude, with the gaps inside the per-subject noise.

8.2 Why the Effect Emerges with Scale

The contrast between the n9 BCIC-2B gap (+0.003, not significant) and the n156 BCIC-2B gap (+0.031, $p = 0.022$) is the most informative comparison in the project. Two readings are consistent with the data. First, an inductive bias helps most once the corpus is large enough that an unstructured baseline would otherwise need to fit montage-specific structure it cannot share, while the geometric encoder shares that structure through g_{ij} . Second, 18 subjects is too few to resolve a ~ 2 -point BAC difference against ~ 2 -point per-subject variance, whereas a larger, more diverse corpus both grows the gap and tightens the estimate. The descriptor ablation supports the first reading, since even the position-only descriptor retains a significant advantage.

8.3 What the Name-Aligned Codex Shows

The codex reaches 0.522 only by name-aligned transfer, and only because all three BCIC-2B names matched the pretraining montage. That makes the comparison conservative twice over: the codex runs at all only by luck of naming, and the one regime it runs in (a PhysioNet-only checkpoint with fully matched names) is tilted in its favor. The channel-independent encoder remains the cleaner baseline for the paired test because, like the geometric encoder, it is montage-agnostic.

9 Conclusion

I implemented and evaluated a geometrically inductive self-supervised EEG encoder: spatial attention conditioned on a geometric descriptor g_{ij} instead of a learned per-electrode codex. I tested the property the proposal called central — zero-shot transfer to an unseen montage — with a leave-one-montage-out, mixed-corpus protocol, comparing against a channel-independent encoder (the only baseline that transfers without modification) and a name-aligned transductive codex, on three open EEG datasets.

The headline result is positive: pretrained on a balanced 156-subject corpus and evaluated zero-shot on the held-out 3-channel BCIC-2B montage, the geometric encoder reaches 0.545 BAC

and beats the channel-independent baseline (0.509) by a statistically significant +0.036 BAC ($p = 0.023$, 95% CI [+0.006, +0.066]). The position-only descriptor also beats it significantly (+0.031, $p = 0.022$, 95% CI [+0.006, +0.056]). The name-aligned codex, in its best case, reaches only 0.522 — below both geometric variants. At small balanced scale the advantage is directionally consistent across all three held-out montages but within noise. Framed as a design study, the project answers its research question — *does geometric conditioning transfer zero-shot to an unseen montage, better than the comparable baseline?* — with a measured yes at sufficient scale, on the montage the available data lets me test at scale, and it names the specific follow-ups that would extend the answer to more montages.

10 Limitations and Future Work

The study’s bounds are mostly data availability, and each points to a concrete follow-up. The items are in roughly decreasing value.

1. **One powered held-out montage.** The significant result is on BCIC-2B only. The other two held-out directions (Sleep-EDFx, PhysioNet MI) can only be tested at the balanced n9 scale, because any balanced corpus containing BCIC-2B is capped at 9 subjects. The most informative follow-up is a second distinct-montage held-out test with enough subjects to be powered — a high-density (≈ 128 -channel) motor-imagery dataset, or the 22-channel BCIC-2A — while training on a large, diverse corpus.
2. **Pretraining scale.** n156 is large for this project but small by foundation-model standards, and the advantage grows from n9 to n156, so this report cannot characterize where it saturates. Pretraining on a larger, more diverse multi-dataset corpus and re-running the probe would show whether the gap keeps growing.
3. **Small cohort, single seed.** BCIC-2B has only 9 subjects, so the headline significance rests on 9 paired LOSO folds. The 95% confidence interval excluding zero guards against this, but gaps of a few BAC points should be read cautiously. Time and compute did not permit multi-seed averaging, so per-seed variance is not reported. Multi-seeding the headline would put a confidence band on the n156 result.
4. **Codex bar is a single favorable point estimate.** The codex is evaluated by name-aligned transfer from a PhysioNet-only checkpoint with all three BCIC-2B names matched — its best case. It carries no paired test, and montages whose names do not match would need a documented fallback before the codex could run. Pretraining a codex on the same mixed 156-subject corpus, with fallbacks (random / nearest-neighbour) for absent names, is the apples-to-apples follow-up.
5. **Descriptor still moderate.** The 10-D g_{ij} captures absolute position, displacement, and distance, but not hemisphere indicators, geodesic distance on a cortical mesh, or anatomy. Richer descriptors — spherical coordinates, an explicit hemisphere indicator, geodesic distance on a cortical-surface mesh — under the fixed-G3 design are future work, building on the ablation showing even position-only transfers.
6. **Cross-session and in-distribution rungs dropped.** This study focuses on cross-montage transfer. The cross-session (Sleep-EDFx night-split) and in-distribution rungs of the earlier ladder are not re-run here, and re-adding them would round out the evaluation.

11 Reused Open-Source Code

- **EEGPT** [14] — github.com/BINE022/EEGPT. I read the EEGPT codebase to validate (a) the cosine ramp for the EMA τ schedule, (b) stop-gradient placement in the alignment loss, and (c) the masked-patch reconstruction convention. Apache 2.0. My code is an independent implementation, not a fork.
- **MNE-Python** [3] — data loading, montage handling, EDF parsing, bandpass filtering, resampling, Sleep-EDFx access.
- **MOABB** [1] — BCIC-2B dataset access (BNCI2014_004). The MOABB dataset registry was also surveyed for future-work montage candidates.
- **PyTorch** — backbone, attention, training loop.
- **scikit-learn** — `StandardScaler`, `LogisticRegression`, balanced accuracy / Cohen’s κ scoring.
- **SciPy** — `scipy.stats` paired t -test, with the 95% confidence interval, for the headline significance test.

The project itself is licensed under the LICENSE file in the public repository at github.com/tianxin-scu/geometric-eeg-ssl.

References

- [1] B. Aristimunya, I. Carrara, P. Guetschel, S. Sedlar, P. Rodrigues, J. Sosulski, D. Narayanan, E. Bjareholt, Q. Barthelémy, R. T. Schirrmeister, R. Kobler, E. Kalunga, L. Darmet, G. Cattani, A. Abdul Hussain, R. Gatti, V. Goncharenko, J. Thielen, T. Moreau, Y. Roy, V. Jayaram, A. Barachant, and S. Chevallier. Mother of all bci benchmarks (moabb), 2025.
- [2] A. M. Bastos and J.-M. Schoffelen. A tutorial review of functional connectivity analysis methods and their interpretational pitfalls. *Frontiers in Systems Neuroscience*, 9, 2016.
- [3] A. Gramfort, M. Luessi, E. Larson, D. A. Engemann, D. Strohmeier, C. Brodbeck, R. Goj, M. Jas, T. Brooks, L. Parkkonen, and M. Hämäläinen. Meg and eeg data analysis with mne-python. *Frontiers in Neuroscience*, 7, 2013.
- [4] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. H. Richemond, E. Buchatskaya, C. Doersch, B. A. Pires, Z. D. Guo, M. G. Azar, et al. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [5] W. L. Hamilton, R. Ying, and J. Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [6] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick. Masked autoencoders are scalable vision learners. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [7] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick. Momentum contrast for unsupervised visual representation learning. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [8] W.-B. Jiang, L.-M. Zhao, and B.-L. Lu. Large brain model for learning generic representations with tremendous eeg data in bci. In *International Conference on Learning Representations (ICLR)*, 2024.
- [9] B. Kemp, A. H. Zwinderman, B. Tuk, H. A. C. Kamphuisen, and J. J. L. Oberyé. Analysis of a sleep-dependent neuronal feedback loop: The slow-wave microcontinuity of the eeg. *IEEE Transactions on Biomedical Engineering*, 47(9), 2000.
- [10] D. Kostas, S. Aroca-Ouellette, and F. Rudzicz. Bendr: Using transformers and a contrastive self-supervised learning task to learn from massive amounts of eeg data. *Frontiers in Human Neuroscience*, 2021.
- [11] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer, and J. R. Wolpaw. Bci2000: A general-purpose brain-computer interface (bci) system. *IEEE Transactions on Biomedical Engineering*, 51(6), 2004.
- [12] M. Tangermann et al. Review of the bci competition iv. *Frontiers in Neuroscience*, 6, 2012.

- [13] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio. Graph attention networks. In *International Conference on Learning Representations (ICLR)*, 2018.
- [14] G. Wang, W. Liu, Y. He, C. Xu, L. Ma, and H. Li. Eegpt: Pretrained transformer for universal and reliable representation of eeg signals. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [15] C. Yang, M. B. Westover, and J. Sun. Biot: Biosignal transformer for cross-data learning in the wild. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.